

Generalisation in theory and practice

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Outline

Quality metrics

- Supervised machine learning problems

Generalisation

Estimating quality

- Methodology

Learning and generalisation

- Introduction

- Bias and variance

- Generalisation

PAC Learning

- The realisable setting

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Supervised machine learning problems

Generalisation

Estimating quality

Methodology

Learning and generalisation

Introduction

Bias and variance

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PAC Learning

The realisable setting

Classification

The classifier as a decision rule

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Randomised predictions given a model $P(y|x)$

Here, we randomly select a class according to our model:

$$\pi(a|x_t) = P(y_t = a|x_t)$$

Accuracy as a classification metric

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The accuracy of a single decision

$$U(a_t, y_t) = \mathbb{I}\{a_t = y_t\} = \begin{cases} 1, & \text{if } a_t = y_t \\ 0, & \text{otherwise} \end{cases}$$

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The accuracy on a dataset

Let $D = \{(x_t, y_t) : t \in [T]\}$ be a dataset. We can measure the accuracy:

$$U(\pi, D) \triangleq \frac{1}{T} \sum_{t=1}^T \pi(y_t | x_t)$$

The expected accuracy of a decision rule

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If $(x, y) \sim P$, the accuracy U of a stochastic decision rule π under the distribution P is the probability it predicts correctly

$$U(\pi, P) \triangleq \int_{\mathcal{X}} dP(x) \sum_{y \in \mathcal{Y}} P(y|x) \pi(y|x)$$

Beyond classification: Generalised decision rules

Consider a spam application, where the e-mail client can decide between different action for emails. Different actions being best for each type of e-mail. The quality of each action can be captured through a utility function.

Utility of the spam decision problem

What utility function would you use for the spam detection problem?

Utility	Pass	Flag	Trash
Normal			
Spam			
Virus			

The utility function $U : \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}$

The utility function $U(y, a)$ is a real-valued function so that, for a label y , we prefer taking action a to a' iff $U(y, a) > U(y, a')$.

The optimal decision

- ▶ A **model** $P(y|x)$ of class probabilities
- ▶ A **utility** $U(y, a)$ for each class and action combination

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We can calculate the expected utility of any decision

$$\mathbb{E}[U|a, x] = \sum_y P(y|x, a)U(y, a) = \sum_y P(y|x)U(y, a)$$

Here the first equality follows from the definition of conditional expectation and $P(y|x, a) = P(y|x)$ as the label does not depend on our actions.

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The optimal decision

For any observation x , and P , we take the action maximising expected utility:

$$a^* = \arg \max_a \mathbb{E}_P[U|a, x]$$

This defines a function $\mathcal{X} \rightarrow \mathcal{A}$, which is the Bayes-optimal decision rule.

The optimal decision rule

- ▶ A **model** $P(y|x)$ of class probabilities
- ▶ A **utility** $U(y, a)$ for each class and action combination
- ▶ A **decision rule** $\pi(a|x)$ assigning probability to action a for every possible input x

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Expected utility over a dataset.

We obtain the expected utility of the **decision rule** by marginalising over all actions

$$U(\pi, D) \triangleq \mathbb{E}[U|\pi, D] \stackrel{D=(x_t, y_t)_{t=1}^T}{=} \sum_{t=1}^T \mathbb{E}[U|\pi, x_t] = \sum_{t=1}^T \sum_{a \in \mathcal{A}} U(y_t, a) \pi(a|x_t)$$

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Expected utility over P

We can marginalise over possible datasets D

$$U(\pi, P) \triangleq \mathbb{E}_P^\pi[U] = \int_{\mathcal{D}} dP(D) U(\pi, D) \stackrel{\text{i.i.d.}}{=} \int_{\mathcal{X}} dP(x) \sum_y P(y|x) \sum_a \pi(a|x) U(y, a)$$

Taking into account the probability

- ▶ For classification, it makes sense to look at the probability of the labels.
- ▶ If we are not very confident about our prediction, this should be taken into account:
- ▶ Define $P(y|x)$ to be our classifier's probability for label y , given features x . Then we can use two simple metrics:

Precision

The average probability of the actual class:

$$\sum_{t=1}^T P(y_t|x_t) / T$$

- ▶ If we always assign probability 1 to the correct label, this score is 1.
- ▶ If we always assign probability $1/m$ to all labels, the score is $1/m$.

Negative Log-Loss

Here we assign look at the **logarithm** of the probability. This really penalises bad guesses.

- ▶ $\sum_{t=1}^T \ln P(y_t|x_t) / T$
- ▶ If we always assign probability 1 to the correct label, this score is 0.
- ▶ If we assign probability 0 to even a single label, the score is $-\infty$.

```
from sklearn.metrics import log_loss
```

in scikitlearn implements log-loss (**not** negative)

Regression

The regressor as a deterministic decision rule

A decision rule π generates a **decision** $a \in \mathbb{R}^m$.

- ▶ For **deterministic** rules $\pi(x)$ is the prediction for x .
- ▶ We need to define the quality of our predictions, either as a utility $U(y_t, a_t)$ (to maximise) or a loss function $\ell(y_t, a_t)$ (to minimise).

Mean-Squared Error Loss on a Dataset

This is the squared difference in predicted versus actual values:

$$\frac{1}{T} \sum_{t=1}^T [y_t - \pi(x_t)]^2$$

Expected MSE

If $(x, y) \sim P$, the expected MSE of a deterministic rule $\pi : \mathcal{X} \rightarrow \mathbb{R}$ is:

$$\mathbb{E}(y - \pi)^2 = \int_{\mathcal{X}} \mathbb{E}[(y - \pi(x))^2 \mid x] dP(x) = \int_{\mathcal{X}} \int_{\mathcal{Y}} dP(x, y) [y - \pi(x)]^2.$$

Probabilistic regression

The regressor as a stochastic decision rule

A decision rule π generates a **decision** $a \in \mathbb{R}^m$.

- ▶ For **stochastic** rules $\pi(a|x)$ defines a density over predictions.
- ▶ In this case it is natural to define $\pi(y_t, x_t)$ as our metric.

Likelihood on a Dataset

The mean-square error is simply the squared difference in predicted versus actual values:

$$\prod_{t=1}^T \pi(y_t|x_t)$$

We will later see a link between this metric, mean-square error and estimation.

Quality metrics

Supervised machine learning problems

Generalisation

Estimating quality

Methodology

Learning and generalisation

Introduction

Bias and variance

Generalisation

PAC Learning

The realisable setting

Training and overfitting

Training data

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The optimal decision rule for P

$$\max_{\pi} U(\pi, P) = \max_{\pi} \int_{\mathcal{X} \times \mathcal{Y}} dP(x, y) \sum_a \pi(a|x) U(a, y).$$

Here we measure performance for the data **distribution** P .

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The optimal decision rule for D

$$\max_{\pi} U(\pi, D) = \max_{\pi} \sum_{(x, y) \in D} \sum_a \pi(a|x) U(a, y).$$

Performance is measured over the **data** D

Generalisation

The fundamental problem

- ▶ We want to maximise $U(\pi, P)$, the **actual** performance.
- ▶ We can only measure $U(\pi, D)$, over **some data**.
- ▶ We have a **learning algorithm** $\lambda : \mathcal{D} \rightarrow \Pi$
- ▶ If $\pi = \lambda(D)$, then we instead measure $U(\lambda(D), D)$.
- ▶ However $\mathbb{E}_P[U(\lambda(D), D)] \geq U(\pi, P) \forall \pi$ (**biased** estimator)

Generalisation

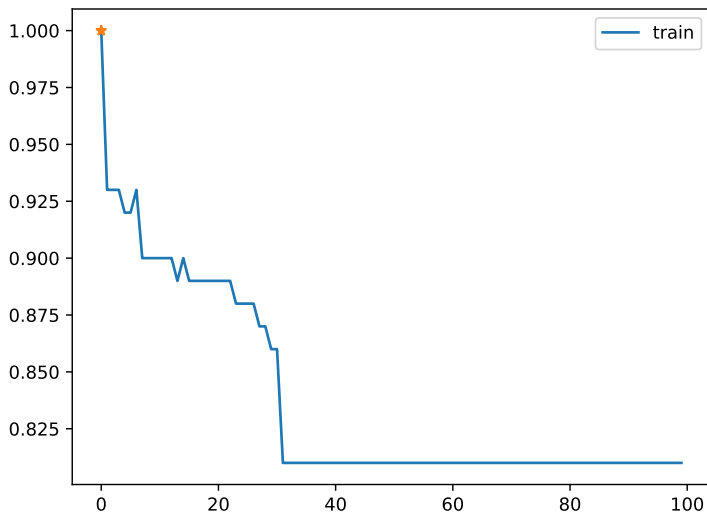
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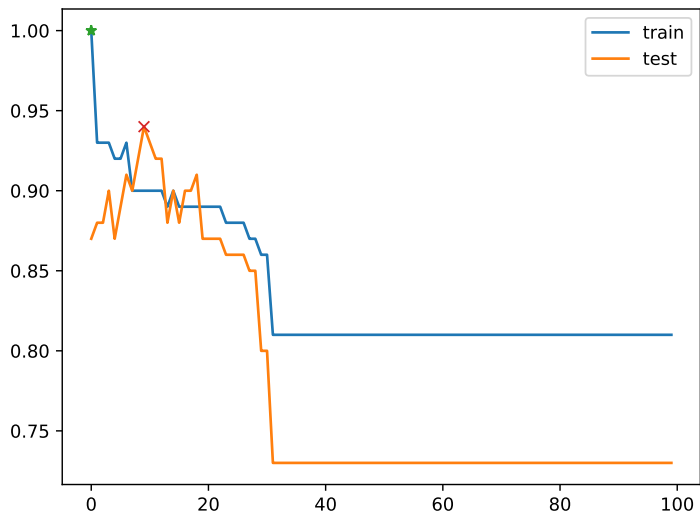
Training and testing

- ▶ Split D in $D_{\text{train}}, D_{\text{test}}$
- ▶ Obtain $\pi = \lambda(D_{\text{train}})$
- ▶ Calculate $U(\pi, D_{\text{test}})$
- ▶ $\mathbb{E}_P[U(\pi, D_{\text{test}})] = U(\pi, P)$ for any fixed π (unbiased estimator)
- ▶ $\mathbb{E}_P[U(\lambda(D_{\text{train}}), D_{\text{train}})] \geq U(\pi, P)$ (biased estimator)

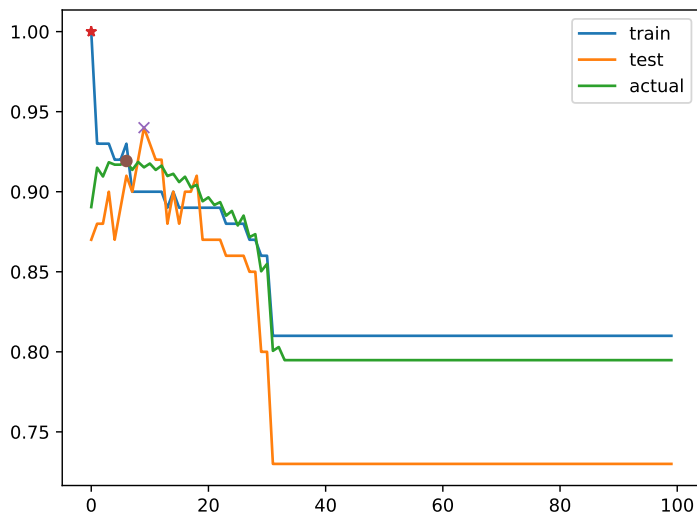
kNN Classifier Accuracy on a single dataset



kNN Classifier Accuracy on a single dataset

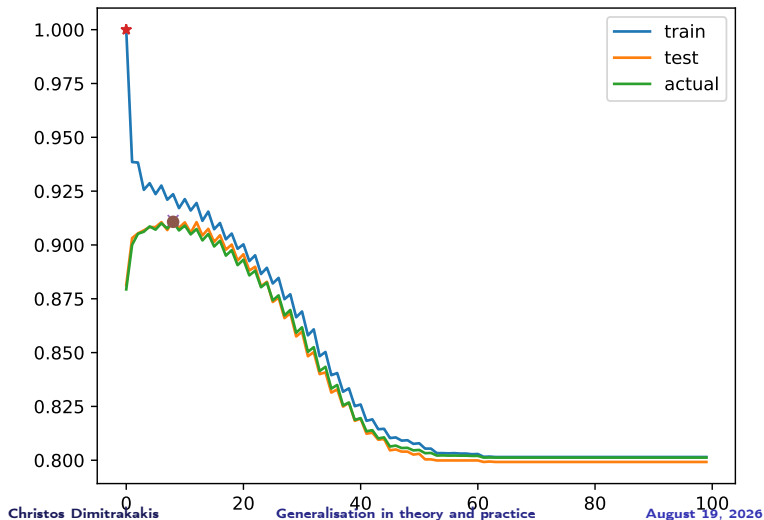


kNN Classifier Accuracy on a single dataset



Expected kNN Classifier Accuracy

Expectation approximated over 100 datasets D sampled from P .



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Supervised machine learning problems

Generalisation

Estimating quality

Methodology

Learning and generalisation

Introduction

Bias and variance

Generalisation

PAC Learning

The realisable setting

The Train/Validation/Test methodology

Main idea

Use each piece of data once to make decisions and measure

Training set

Use to decide low-level model parameters

Validation set

Use to decide between:

- ▶ different hyperparameters (e.g. K in nearest neighbours)
- ▶ model (e.g. neural networks versus kNN)

Test set

Use to measure the final quality of a model

Cross-validation (XV)

Idea

- ▶ Use XV to select hyperparameters instead of a single train/valid test.

Methodology

- ▶ Split training set D in k different subsets
- ▶ At iteration i
- ▶ Use the i -th subset for validation
- ▶ Use all the remaining $k - 1$ subsets for training
- ▶ Average results on validation sets

Cross-validation example: A mean estimate

Live coding in Python:

- ▶ Get a mean estimate
- ▶ Perform cross-validation

Hand-crafted

```
def xv_shuffler(x, estimator, scoring, n_folds):  
    rng = np.random.default_rng  
    # shuffle data  
    T = len(x)  
    indices = np.arange(T)  
    shuffle(indices)  
    fold_size = np.ceil(T / n_folds)  
    fold_start = np.zeros(n_folds)  
    fold_end = np.zeros(n_folds)  
    # create folds  
    for k in range(n_folds):  
        fold_start = k * fold_size  
        fold_end = np.min((k+1) * fold_size, T)  
    # for each fold:
```

1. Run estimator on $k+1$ folds

Bootstrapping

- ▶ Express uncertainty by resampling the data.
- ▶ Repeat your calculations for each resample

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BootstrapSample(D)

```
input Data  $D = (z_1, \dots, z_T)$ , of size  $T$   
for  $t \in \{1, \dots, T\}$  do  
    Select  $i$  uniformly in  $[T]$   
    Add the  $i$ -th point to  $D_b$   
end for  
return  $D_b$ 
```

Bootstrapping

- ▶ Express uncertainty by resampling the data.
- ▶ Repeat your calculations for each resample

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for $t \in \{1, \dots, T\}$ **do**
 Select i uniformly in $[T]$
 Add the i -th point to D_b
end for
return D_b

BootstrapEstimate(D, λ, N)

input Data $D \in \mathcal{D}$, algorithm $\lambda : \mathcal{D} \rightarrow \Theta$, $N > 0$ number of samples
for $n \in \{1, \dots, N\}$ **do**
 $\theta_n = \lambda(\text{BootstrapSample}(D))$
end for
return $\{\theta_n : n \in [N]\}$

The wrong way to do XV for subset selection

1. Screen the predictors: find a subset of “good” predictors that show fairly strong (univariate) correlation with the class labels.
2. Using just this subset of predictors, build a multivariate classifier.
3. Use cross-validation to estimate the unknown tuning parameters and to estimate the prediction error of the final model.

Is this a correct application of cross-validation?

Consider a scenario with $N = 50$ samples in two equal-sized classes, and $p = 5000$ quantitative predictors (standard Gaussian) that are independent of the class labels. The true (test) error rate of any classifier is 50%.

The right way to do XV for feature selection

1. Divide the samples into K cross-validation folds (groups) at random.
2. For each fold $k = 1, 2, \dots, K$
3. Find a subset of “good” predictors that show fairly strong (univariate) correlation with the class labels, using all of the samples except those in fold k .
4. Using just this subset of predictors, build a multivariate classifier, using all of the samples except those in fold k .
5. Use the classifier to predict the class labels for the samples in fold k .

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Level notes

- **BSc:** The point isn't the procedure itself: it's noticing that cross-validation can silently give you a fake guarantee if information from outside a fold leaked into the model-building step before validation happened. Spotting this kind of methodological trap is a core scientific-method skill, and one you should actively check for in your own project.
- **MSc:** **Why** does the wrong version give a near-zero apparent error despite a true error of 50%? Connect it to the estimation-error bound above: what is $|\hat{\Pi}|$ effectively doing once you've screened 5000 predictors first?

Lab

- ▶ scikitlearn KNN
- ▶ scikitlearn performance measures
- ▶ train/test/validate plot with increasing k
- ▶ XV plot with increasing k
- ▶ Scaling and other preprocessing
- ▶ Effect of outliers on preprocessing
- ▶ Optional: Bootstrap performance evaluation

Quality metrics

Supervised machine learning problems

Generalisation

Estimating quality

Methodology

Learning and generalisation

Introduction

Bias and variance

Generalisation

PAC Learning

The realisable setting

Learning and generalisation

How well can decision rule perform?

Estimation theory view

- ▶ Bias: The **expected difference** between the estimated value and the unknown parameter
- ▶ Variance: Just the **variance** of that estimator, as a random variable.
- ▶ Related to generalisation error.

Learning theory view

- ▶ Approximation ability: How well a **class of rules** can approximate the **optimal** one.
- ▶ Statistical error: How **easy** it is to choose the best rule in the class from **data**
- ▶ Directly bounds the generalisation error.

Unbiased estimators

Definition (Estimator)

An estimator is a function $f : \mathcal{D} \rightarrow \Theta$, where Θ is a set of parameters. For any given dataset $D \in \mathcal{D}$, it returns a single estimate $\hat{\theta} = f(D)$.

Definition (Unbiased estimator)

An estimator is **unbiased** if, for any θ and corresponding distribution $P(D|\theta)$:

$$\mathbb{E}[f \mid \theta] = \sum_D f(D)P(D|\theta) = \theta.$$

Example (Sample mean estimator)

Consider $D = (x_1, \dots, x_T)$ with $x_t \sim P$ being i.i.d samples with $\mathbb{E}[x_t] = \theta$. The sample mean estimator $f(D) = \sum_t x_t / T$ is unbiased, as :

$$\mathbb{E}[f] = \mathbb{E} \left[\sum_{t=1}^T x_t / T \right] = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[x_t] = \frac{1}{T} \sum_{t=1}^T \theta = \theta.$$

Example of a biased and unbiased estimator: Training error

- ▶ $U(\pi, D)$ is the measured accuracy of a classifier π on D
- ▶ $U(\pi, P) = \mathbb{E}_{D \sim P}[U(\pi, D)]$ is the actual accuracy. So $U(\pi, D)$ is unbiased.
- ▶ $\lambda(D) = \arg \max_{\pi} U(\pi, D)$ is a learning algorithm picking the best classifier for a dataset D .
- ▶ Then $U(\lambda(D), D)$ is biased, as for any π'

$$\mathbb{E}_{D \sim P}[U(\lambda(D), D)] = \int_D dP(D) U(\lambda(D), D) \quad (1)$$

$$= \int_D dP(D) \max_{\pi} U(\pi, D) \quad (2)$$

$$\geq \int_D dP(D) U(\pi', D) \quad (3)$$

$$= U(\pi', P) \quad (4)$$

i.e. the expected value of the training accuracy is higher than the accuracy of **any** classifier.

The bias/variance trade-off

This is the classical statistics formulation, applied to a regression estimator f_D , not generally for $U(\pi, P)$. It's good background, and the proof below is worth doing for the practice, but it's **optional**: the regret decomposition already covers what you need to know about generalisation error in this course.

Bias/Variance tradeoff

- ▶ Dataset $D \sim P$ (the dataset $D = \{(x_t, y_t) | t = 1, \dots, T\}$ is sampled from the unknown distribution P)
- ▶ Predictor $f_D(x)$ (it makes predictions about individual x)
- ▶ Target function $y = f(x) + \epsilon$ (we want $f_D \approx f$).
- ▶ $\mathbb{E}(\epsilon) = 0$ zero-mean noise with variance $\sigma^2 = \mathbb{V}(\epsilon)$

MSE decomposition

$$\mathbb{E}[(f - f_D)^2] = \mathbb{V}(f_D) + \mathbb{B}(f_D)^2 + \sigma^2$$

Variance

How sensitive the estimator is to the data

$$\mathbb{V}(f_D) = \mathbb{E}[(f_D - \mathbb{E}(f_D))^2]$$

Bias

What is the expected deviation from the true function

$$\mathbb{B}(f_D) \triangleq \mathbb{E}[(f_D - f)]$$

Example: mean estimation

- ▶ Data $D = y_1, \dots, y_T$ with $\mathbb{E}[y_t] = \mu$.
- ▶ Goal: estimate μ with some estimator f_D to minimise
- ▶ MSE: $\mathbb{E}[(y - f_D)^2]$, the expected square difference between new samples our guess.

Optimal estimate

To minimise the MSE, we use $f^* = \mu$. This gives us two ideas:

Empirical mean estimator:

- ▶ $f_D = \sum_{t=1}^T x_t / T$.
- ▶ $\mathbb{V}(f_D) = \mathbb{E}[f_D - \mu]^2 = \sigma^2 / T$
- ▶ $\mathbb{B}(f_D) = 0$. (unbiased estimator)

Laplace mean estimator:

- ▶ $f_D = \sum_{t=1}^T (\lambda + x_t) / T$.
- ▶ $\mathbb{V}(f_D) = \mathbb{E}[f_D - \mu] = \frac{1}{1 + \sqrt{T}}$
- ▶ $\mathbb{B}(f_D) = O(1/T)$.

A proof of the bias/variance trade-off

- ▶ RV's $y_t \sim P$, $\mathbb{E}[y_t] = \mu$, $y_t = \mu + \epsilon_t$.
- ▶ Estimator f_D , $D = y_1, \dots, y_{t-1}$.

$$\begin{aligned}
 \mathbb{E}[(f_D - y_t)^2] &= \mathbb{E}[f_D^2] - 2\mathbb{E}[f_D y_t] + \mathbb{E}[y_t^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D y_t] + \mathbb{E}[y_t^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D]\mathbb{E}[y_t] + \mathbb{E}[y_t^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D]\mu + \mathbb{E}[y_t^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D]\mu + \mathbb{E}[(\mu + \epsilon_t)^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D]\mu + \mathbb{E}[\mu^2 + 2\mu\epsilon_t + \epsilon_t^2] \\
 &= \mathbb{V}[f_D] + \mathbb{E}[f_D]^2 - 2\mathbb{E}[f_D]\mu + \mu^2 + \sigma^2 \\
 &= \mathbb{V}[f_D] + (\mathbb{E}[f_D] - \mu)^2 + \sigma^2 \\
 &= \mathbb{V}(f_D) + \mathbb{B}(f_D)^2 + \sigma^2
 \end{aligned}$$

Level notes

- ▶ **BSc**: Optional — skip unless you want the algebra practice.
- ▶ **MSc**: A clean derivation to be able to reproduce; every line uses only linearity

Generalisation error

Regret decomposition

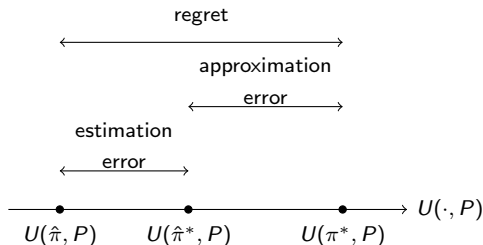
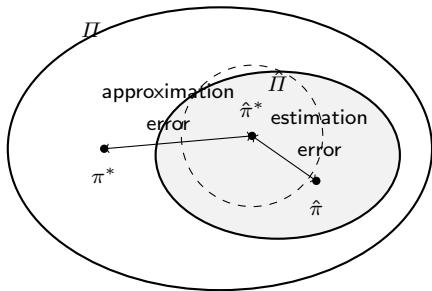
Let the optimal rule be $\pi^* \in \Pi$, the best approximate rule be $\hat{\pi}^* \in \hat{\Pi}$ and our rule be $\hat{\pi} \in \hat{\Pi}$. We call the difference between the performance of π^* and $\hat{\pi}$ our **regret**:

$$\underbrace{U(\pi^*, P) - U(\hat{\pi}, P)}_{\text{regret}} = \underbrace{U(\pi^*, P) - U(\hat{\pi}^*, P)}_{\text{approximation error}} + \underbrace{U(\hat{\pi}^*, P) - U(\hat{\pi}, P)}_{\text{estimation error}}$$

We can bound the regret by bounding each term separately.

- ▶ The **approximation error** tells us how expressive our class of rules is, i.e. how much we lose by looking at a restricted class $\hat{\Pi}$ of rules. It is similar to estimator **bias**.
- ▶ The **statistical error** tells us how well the empirical performance on D approximates the true performance. It is similar to estimator **variance**.
- ▶ As a rule of thumb, the larger our class $\hat{\Pi}$, the better the possible approximation but the higher the statistical error.

Geometric intuition



- ▶ The fact that $\hat{\Pi} \neq \Pi$ creates the **approximation error**.
- ▶ π is a function of dataset D , which is random. This noise results in **estimation error** (dashed circle: where $\hat{\pi}$ could land, over different draws of D).
- ▶ Both panels show the **same** three rules. On the left, the gaps come from *where* the rules sit (inside or outside $\hat{\Pi}$, and where exactly within it). On the right, the same gaps are measured in the only unit that actually matters: utility.

Level notes

- ▶ **BSc**: Don't worry about reproducing this diagram — just be able to say, in your own words, which gap is "the model's fault" and which is "bad luck".

Approximation error

- ▶ Our model limits us to a set of decision rules $\hat{\Pi} \neq \Pi$.
- ▶ The most we could do is find the best rule in $\hat{\Pi}$.
- ▶ This still leaves a gap:

$$\Delta \triangleq \max_{\pi \in \Pi} U(\pi, P) - \max_{\hat{\pi} \in \hat{\Pi}} U(\hat{\pi}, P)$$

The gap can be characterised in some cases.

Example: ϵ -net on Lipschitz $U(\cdot, P)$.

- ▶ Assume $U(\pi, P)$ is a Lipschitz function of π for all P , i.e. $|U(\pi, P) - U(\pi', P)| \leq Ld(\pi, \pi')$ for some metric d .
- ▶ Let $\hat{\Pi}$ be an ϵ -net on Π , i.e. $\max_{\pi \in \Pi} \min_{\pi' \in \hat{\Pi}} d(\pi, \pi') = \epsilon$.
- ▶ Then $\Delta \leq L\epsilon$.

Estimation error

- ▶ First, let us bound $U(\hat{\pi}^*, P) - U(\hat{\pi}, P)$ by making an assumption.
- ▶ Then, we can prove that our assumption holds with high probability.

Lemma

Let $f, g : S \rightarrow \mathbb{R}$. If $\|f - g\|_{\infty} \leq \epsilon$ and $f(x) \geq f(z)$, while $g(y) \geq g(z)$, for all z , i.e. x, y maximise f, g respectively

$$f(x) - f(y) \leq 2\epsilon.$$

This holds as: $f(x) - f(y) \leq g(x) + \epsilon - f(y) \leq g(y) + \epsilon - f(y) \leq 2\epsilon$.

Corollary

If $|U(\pi, P) - U(\pi, D)| \leq \epsilon$ for all π then

$$U(\hat{\pi}^*, P) - U(\hat{\pi}, P) \leq 2\epsilon$$

- ▶ Let us now prove that, with high probability, $|U(\pi, P) - U(\pi, D)| \leq \epsilon$.

Bounding the estimation error

For any fixed rule $\pi \in \Pi$ and utility function $U : \Pi \times \mathcal{X}^T \rightarrow [0, 1]$,

$$P^T(|U(\pi, D) - U(\pi, P)| \geq \epsilon) \leq 2 \exp(-2T\epsilon^2).$$

This is a direct application of Hoeffding's inequality¹. Taking the union bound over the set $\hat{\Pi}$ gives:

$$P^T(\exists \pi \in \hat{\Pi} : |U(\pi, D) - U(\pi, P)| \geq \epsilon) \leq 2|\hat{\Pi}| \exp(-2T\epsilon^2).$$

Setting the right side equal to δ and re-arranging,

$$P^T \left(\max_{\pi \in \hat{\Pi}} |U(\pi, D) - U(\pi, P)| \geq \sqrt{\frac{\ln(2|\hat{\Pi}|/\delta)}{2T}} \right) \leq \delta.$$

Example: ϵ -net.

In a n dimensional space we require $|\hat{\Pi}| = O(\epsilon^{-n})$. This means that our statistical error is $O(\sqrt{n \ln(1/\epsilon\delta)/T})$.

¹See Hoeffding's inequality in the confidence intervals presentation 

The finite hypothesis algorithm

- ▶ Input: a finite set of rules $\hat{\Pi}$, data D , utility U
- ▶ Return $\hat{\pi} \in \arg \max_{\pi \in \hat{\Pi}} U(\pi, D)$.

Regret of the finite hypothesis algorithm.

With probability $1 - \delta$

$$U(\hat{\pi}, P) \geq U(\hat{\pi}^*, P) - \sqrt{2 \ln(2|\hat{\Pi}|/\delta)/T} \quad (5)$$

$$U(\pi^*, P) - U(\hat{\pi}, P) \leq \Delta + \sqrt{2 \ln(2|\hat{\Pi}|/\delta)/T} \quad (6)$$

Examples

- ▶ ML estimation: $U(\beta, D) = P_\beta(D)$ is the data likelihood.
- ▶ Accuracy, etc: $U(\pi, D)$.

VC Dimension

Here we consider sets Π of deterministic rules $\pi : \mathcal{X} \rightarrow \{0, 1\}$.

Shattering

If a $S \subset \mathcal{X}$ can with $|S| = m$, can be assigned any labelling $y_1, \dots, y_m$ by a $\pi \in \Pi$, then we say Π shatters S .

The VC dimension

This is the largest-size set S that Π can shatter.

Example: Perceptrons on $\mathbb{R}^2$

This class has VC dimension 3 on the plane.

Quality metrics

Supervised machine learning problems

Generalisation

Estimating quality

Methodology

Learning and generalisation

Introduction

Bias and variance

Generalisation

PAC Learning

The realisable setting

Binary classification

Learning algorithm λ

- ▶ Takes data $D = \{(x_t, y_t)\}$ as input
- ▶ Generates deterministic decision rules $\pi : X \rightarrow \{0, 1\}$,

The loss of a rule π .

- ▶ Assume an existing concept class $\pi^* \in \Pi$
- ▶ Distribution $x_t \sim P$ is i.i.d. and $x_1, \dots, x_T \sim P^T$.
- ▶ The loss under distribution P is

$$L(\pi) = P(\{x : \pi(x) \neq \pi^*(x)\})$$

Realisable PAC learner

- ▶ $\lambda : (\mathcal{X} \times \mathcal{Y})^* \rightarrow \Pi$ is (ϵ, δ) -PAC, if for any P and $\epsilon, \delta > 0$, and any concept $\pi^* \in \Pi$, there is T such that

$$P^T(\{D : L[\lambda(D)] > \epsilon\}) < \delta, \quad D = (\{x_t, \pi^*(x_t)\}), x_t \sim P.$$